

AAYUSH KUMAR

+91 9204362088 | aayush2615@gmail.com | [linkedin.com/in/aayush2724](https://www.linkedin.com/in/aayush2724) | github.com/aayush2724 | aayush-2724.vercel.app

Education

The National Institute of Engineering

Bachelor of Engineering in Computer Science and Engineering (AI & ML)

Aug 2024 – May 2028

CGPA: 9.3/10

Technical Skills

Languages: C++, JavaScript (ES6+), TypeScript, Python, SQL, Java

Frontend: React.js, Vite, Tailwind CSS, HTML5, CSS3, Framer Motion

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, WebSockets, Server-Sent Events (SSE), BullMQ

Databases: PostgreSQL, MongoDB, Redis, Firebase

DevOps & Tools: Docker, Linux (Fedora), Git, GitHub, GitHub Actions, CI/CD, Postman, Vercel, Render

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Projects

Auralis

Jul 2026

React.js, TypeScript, FastAPI, LangGraph, PostgreSQL, Redis, FAISS, Docker

[GitHub](#) | [Live](#)

- Built an **AI-powered sales intelligence platform** using LangGraph that classifies objections with **94% confidence**, detects personas and sentiment, and generates contextual responses in under 2 seconds.
- Implemented a **RAG pipeline** with FAISS and a production-ready stack using React, FastAPI, PostgreSQL, Redis, JWT, WebSockets, Docker, and GitHub Actions CI/CD.
- Designed a modular **multi-agent architecture** with strategy-specific routing, Redis-backed session management, analytics tracking, and scalable deployment on Vercel and Render.

Beatzy

Apr 2026

Python, FastAPI, React.js, Node.js, PostgreSQL, Redis, BullMQ, HuggingFace

[GitHub](#) | [Live](#)

- Developed a full-stack **music intelligence platform** supporting BPM, key, chord, mood analysis, lyric synchronization, and metadata enrichment using librosa and YAMNet.
- Designed an asynchronous processing pipeline with **BullMQ, Redis, and PostgreSQL**, with microservices deployed on Render and Vercel.
- Integrated external music metadata services and optimized repeated analyses using **Redis caching** to reduce processing latency.

AlgoVision

Sep 2026

JavaScript, FastAPI, Python, Three.js, GSAP, SVG, Docker

[GitHub](#) | [Live](#)

- Built an interactive **DSA learning engine** with **22 live algorithm tracers**, including Dijkstra, BFS/DFS, MSTs, sorting, binary search, heaps, sliding window, and dynamic programming.
- Implemented **step-by-step execution with forward/backward scrubbing**, live operation counters, complexity analysis, and side-by-side algorithm comparison using real trace data.
- Developed a **FastAPI backend and AI microservice** for algorithm detection and contextual explanations, with an offline-first browser emulator and Docker-based deployment.

Experience

OWASP NIE Student Chapter

Aug 2025 – Present

President

NIE, Mysore

- Lead the chapter's **technical initiatives, developer events, and security-focused programs**, coordinating student teams and supporting 100+ participants across campus events.
- Organize and deliver technical workshops on **Git, GitHub, open source, web development, and cybersecurity**, helping students build practical development skills.
- Oversee technical infrastructure, event workflows, and collaborative development processes while mentoring members across the chapter's technical teams.

Open Source Contributor

2025 – Present

Independent

[GitHub](#)

- Build and maintain **production-grade full-stack and ML applications** spanning React, Node.js, FastAPI, PostgreSQL, Redis, and modern AI/ML pipelines.
- Maintain deployment-ready repositories with **Docker, GitHub Actions, and CI/CD workflows**, emphasizing reproducibility, documentation, and scalable architecture.

Achievements

- **Hackathons:** **3× Hackathon Finalist**; secured **3rd Place** at ThinkRoot x Vortex'26 (NIT Trichy), competing against 100+ teams and building [LeadForge](#) in 24 hours.
- **Competitive Programming:** Solved **600+ DSA problems** across platforms, covering Graphs, Dynamic Programming, Trees, and Advanced Data Structures.
- **Open Source:** Maintain a portfolio of **full-stack and ML applications** with production-oriented architecture, Docker-based environments, and CI/CD workflows.